UE23CS320A – Capstone Project Approval

Project Title: Echo – Your AI Memory Logger

Project ID: 92

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Outline

- Problem Statement
- Scope and Feasibility study
- Applications/Use cases
- Expected Deliverables
- Capstone (Phase-I Phase-II, Phase-III) Project Timeline
- Any other information

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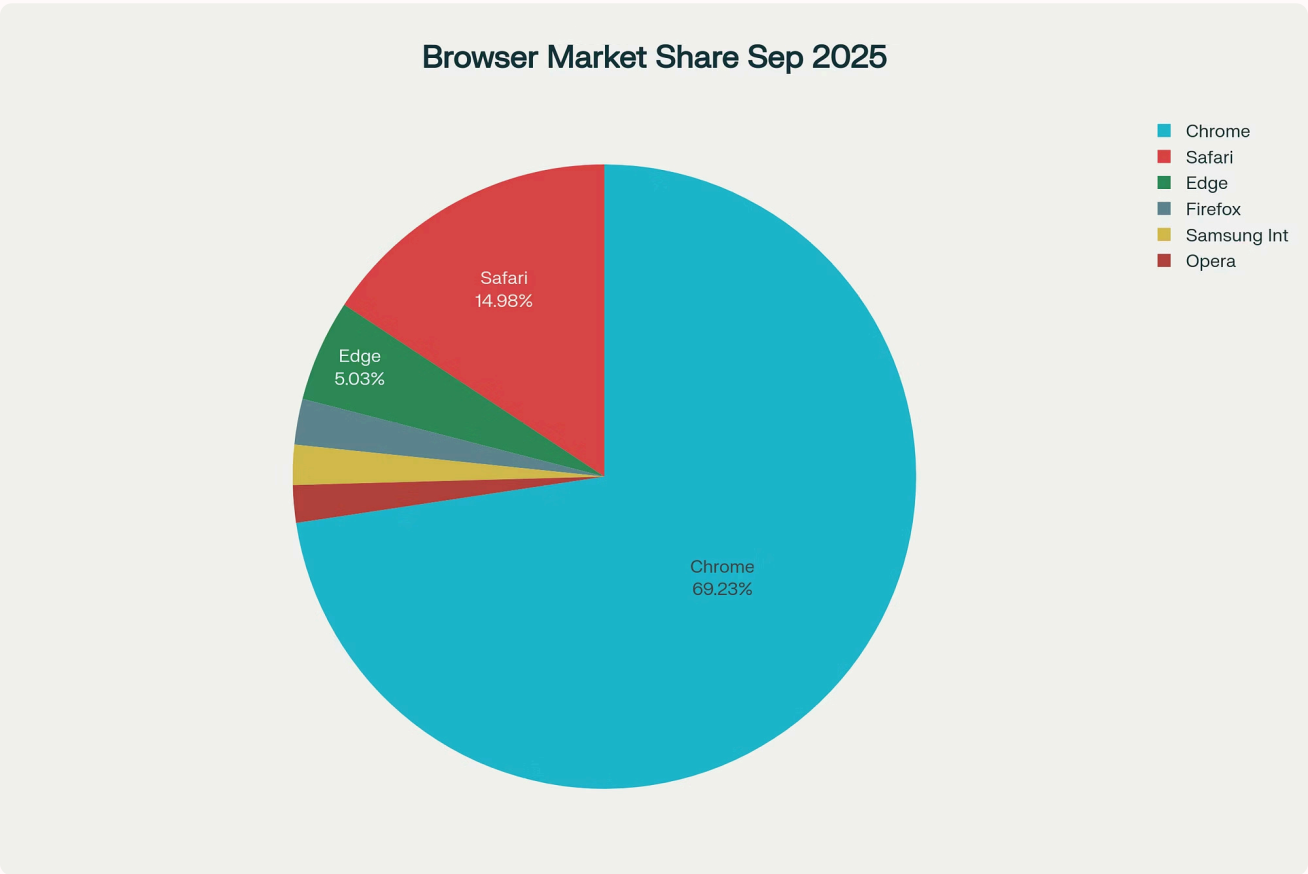
Problem Statement:

Navigating the digital landscape is like searching for a needle in a haystack—information overload hinders us from uncovering what truly counts.

- The average internet user visits between **5 and 20+ unique websites daily**.
- People forget **up to 70% of new information within 24 hours**.
- Knowledge workers spend **1.8–2.5 hours each day searching for information they've already seen**.
- Employees can spend up to **30% of their workday searching for and gathering information**.
- Professionals typically switch between **35 different websites or platforms per day**.
- After just one week, people retain **less than 25% of newly learned information**

Why does this happen? Because our digital ecosystem is fragmented. Critical knowledge is split across Gmail threads, Chrome's sprawling browsing history, and countless YouTube videos. There is no single memory. Instead, there are scattered islands—and precious time is lost trying to reconnect them.



Chrome dominates the web with over **3.98 billion users** and a **69% global market share**.

But most users face- the same digital struggle: Repeating clicks to open tabs already visited, wasting time even when the right page is buried in their history.

The "grain in the husk" problem: Trying to recall **one specific page** from a **sea of millions**, users often give up even though the answer is in their own data.

Result: Productivity is lost, key information is hidden, digital clutter overwhelms, and frustration grows.

And this problem isn't just limited to browsing. When an important email arrives, users scroll through endless lists or clumsy search results, missing crucial information at critical moments. On YouTube, the cycle repeats: after hours of careful research and curated playlists, the video that sparked fresh insight is lost among hundreds, impossible to recall without endless scrolling and guesswork.

> **Manual history and search across these platforms are fundamentally broken—outdated, overwhelming, and ignored by most users.** In an era of digital abundance, our memory is more scattered—and more precious—than ever.

Sources: arXiv | TalentCards | The Learning Guild | Wikipedia | Cottrill Research | Unleash | ProProf | BloggingWizard | LearnUpon

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Introducing Echo: Your AI Memory Logger

Scope:

Unified Recall:

Echo automatically gathers your Chrome history, Gmail, and YouTube activity into a single, searchable digital memory—no more checking each one separately. ~Novelty

Instant, AI-Powered Search:

Find any website, email, or video you've seen just by asking a simple, natural-language question. No special keywords or date filters needed. ~ Novelty

Context-Aware Results:

Even vague prompts like "that internship link from last Friday" or "email about Kahoot code" bring up exactly what you need.

One-Click AI Summaries:

With a single click, Echo instantly summarizes the webpage or video you're viewing—no need to copy-paste into an external chatbot or summary tool.

Activity Analytics:

Echo breaks down your digital time by category—study, work, entertainment, and more—showing you at a glance how you spend your online hours. On user request, it can provide even finer separation (e.g., "work" into exam preparation vs. project research) to help you track exact time spent on each. ~ Novelty

All-in-One Chrome Dashboard:

Access unified search, AI memories, summaries, and activity trends right from your Chrome homepage—no need to jump between apps or tabs.

Regret Button & Proactive Nudges:

Mark regrets (like wasted YouTube binges) and Echo spots repeat habits, gently reminding you before you slip again. ~Novelty

Personal Wellness Guidance:

Receive smart reminders and tips to build better study, work, and relaxation routines—tailored to your real digital behavior. ~Novelty

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Feasibility & Technical Foundation

Echo is built on a robust technical framework, leveraging modern technologies and privacy-first design.

Technical Strengths

- **Integrated APIs:** Stable APIs for Chrome, Gmail, YouTube, Calendar, and Drive.
- **Modern Tech Stack:** React/Next.js (UI), Node.js/Express (Backend), Firebase/Firestore (Data Storage).
- **Advanced AI/ML:** LLM APIs (OpenAI/Claude), TensorFlow, PyTorch for conversational and analytic models.
- **Modular Architecture:** Microservices for data retrieval, AI processing, and user interaction.

Key Challenges

- Cohesive mapping and synchronization from multi-source APIs.
- Building empowering yet unobtrusive analytics.
- Maintaining user privacy, high security, and responsive performance at scale.

Privacy & Compliance: Privacy-first by design, with local storage, explicit opt-in cloud sync, user controls, and GDPR alignment.

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Applications/Use cases

Students:

- Find watched lectures, research articles, and assignment emails instantly.
- Track and visualize true study time.
- Summarize any webpage or video for quick revision.

Professionals:

- Instantly retrieve project mails, meeting notes, and reference sites.
- Summarize work content at a glance.
- Analyze workflow to optimize productivity.

General Users:

- Recall key recipes, landlord emails, or helpful videos.
- See time split between work, entertainment, and more.
- Get wellbeing alerts and weekly digital habit summaries.

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Expected Deliverables

01

Capstone-I: Foundation

Literature review, UI prototype, data flow/security design, initial privacy testing, first POC with basic history/email ingest.

02

Capstone-II: Core Features

Working unified memory system (Chrome+Gmail+YouTube), LLM-powered recall & summarization, initial analytics/behavior tracking, documentation.

03

Capstone-III: Advanced & Deployment

Full-featured Chrome extension with regret/reminder, cross-device sync, deployed system, user pilot results, final technical report.

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Capstone Project Timeline

1

Phase I: Foundation (Aug 2025 – Jan 2026)

- Research documentation on Google APIs and integration capabilities
- System architecture and privacy framework design
- Prototype concepts for core Echo features (basic Chrome extension, initial Gmail API setup)
- Initial LLM (large language model) integration
- Feasibility analysis and initial testing
- Early technical documentation

2

Phase II: Core Features (Feb 2026 – Jul 2026)

- Development of unified memory retrieval system (browsing, email, video integration)
- Full setup of Gmail and YouTube API integrations with working proof-of-concept workflows
- Implementation of natural language interface and in-place summarization
- Agentic AI features introduction (smart reminders, draft support)
- User interface refinement with analytics and summaries
- Progress documentation and evaluation

3

Phase III: Advanced & Deployment (Aug 2026 – Jan 2027)

- Complete, tested version of Echo with advanced AI workflows and memory features
- Cross-device synchronization and wellness analytics
- Performance, security, and privacy optimization
- User testing, validation, and deployment
- Comprehensive user documentation and final technical report

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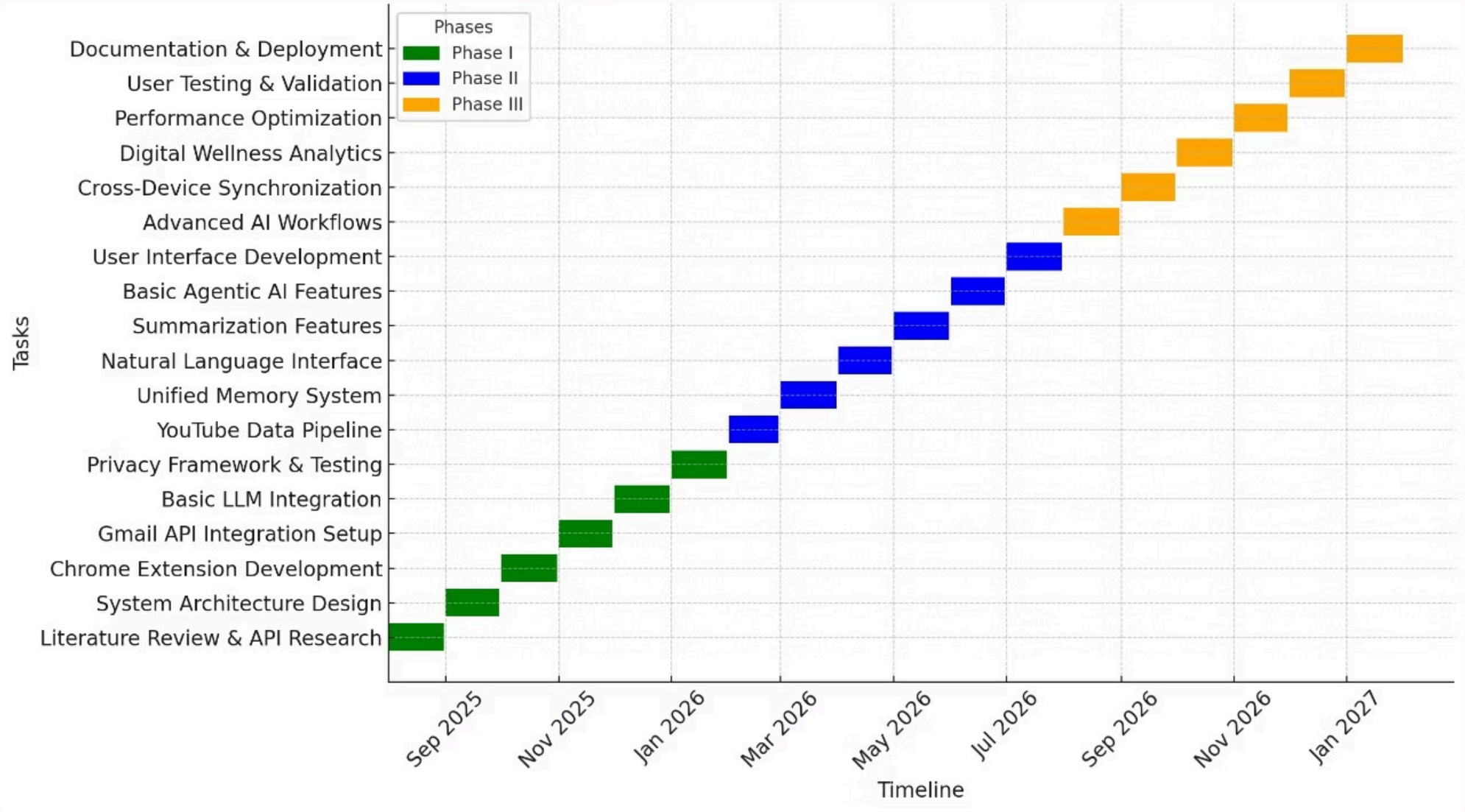
PROJECT TIMELINE - GANTT CHART



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Echo: Your AI Memory Logger Buddy

Echo is more than a productivity tool; it’s a personal companion. For students, it reclaims lost time spent searching. For professionals, it ensures instant access to critical information. For everyone, it tackles digital distractions, offering smart reminders, actionable insights, and a unique "regret button" to foster healthier habits.

> **Echo: Because your digital memory deserves to be more than just a grain lost in the husk.**

Empowering you with clarity, healthier habits, and peace of mind.

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